# Software Requirements Specification (SRS)

## Product: Private Real-Time AI Meeting Coach

## Codename: GleaMeet

## Version: 1.0

## Status: Draft for Engineering Handoff

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# 1. Document Purpose

This document specifies the functional, non-functional, architectural, data, privacy, security, and acceptance requirements for a **private real-time AI meeting coach** that operates first as a **Google Meet browser extension** and later expands to **Microsoft Teams**, **Zoom**, and **Slack**.

The product provides **private, real-time, behavioral-science-grounded coaching** to a single meeting participant. All coaching must be grounded in a versioned **Behavioral Law Registry**. The initial registry is based on selected laws operationalized from:

- Daniel Kahneman
- Robert Cialdini
- Richard Thaler

This SRS is written to be directly usable by a coding agent or implementation team.

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# 2. Product Summary

### 2.1 Product Goal

Help a meeting participant improve their performance **during a live meeting** by delivering private, timely, low-latency prompts such as:

- “Pause. Ask one clarifying question.”
- “Acknowledge their concern first.”
- “You may be anchoring too early.”
- “State the tradeoff, not just the downside.”
- “Assign owner and next step.”

### 2.2 Product Principles

The system must be:

- private by default
- secure by design
- explainable
- grounded in observable evidence
- behavior-focused, not identity-focused
- useful in real time with low cognitive load
- extensible to multiple meeting platforms

### 2.3 Core Product Positioning

This is a **personal meeting coach**, not a surveillance product.

The product is for the individual user only. It is not for:

- employer monitoring
- manager dashboards
- team ranking
- personality assessment
- deception detection
- psychological diagnosis

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## 3. Scope

### 3.1 In Scope for v1

- Google Meet browser extension
- explicit per-meeting consent
- real-time in-call coaching prompts
- post-meeting recap
- law-registry-driven prompting
- low-latency signal extraction
- user-specific coaching only

- account creation and authentication
- configurable retention and deletion controls
- event logging for explainability and debugging

### 3.2 Out of Scope for v1

- live audio whisper coaching
- AI-generated speech on behalf of the user
- autonomous meeting participation
- organizational dashboards
- manager access to coaching data
- mobile apps
- Teams, Zoom, and Slack production support
- cross-user benchmarking
- psychometric profiling
- sentiment diagnosis as a personal trait
- covert recording or invisible capture

### 3.3 Future Scope

- Teams adapter
- Zoom adapter
- Slack huddle/call adapter
- pre-meeting preparation
- meeting-goal-aware prompting
- personalized coaching models
- calendar integration
- richer trend analytics

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## 4. Users and Stakeholders

### 4.1 Primary User

An individual professional who wants to improve live meeting behavior.

Examples:

- founders
- executives
- product managers
- sellers
- consultants

- managers
- interview candidates

### 4.2 Secondary Stakeholders

Internal only:

- engineering
- product
- design
- security
- behavioral science curation team
- QA

No external viewer role exists in v1.

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## 5. Definitions

### 5.1 Meeting Coach

The full product system.

### 5.2 Platform Adapter

Platform-specific code that detects meetings, captures supported signals, and normalizes them into a shared event schema.

### 5.3 Behavioral Law Registry

A versioned registry of operationalized behavioral laws with:

- law metadata
- observable inputs
- trigger logic
- disconfirming logic
- confidence requirements
- allowed prompt templates
- risk notes

### 5.4 Observable Signal

A measurable conversational or linguistic feature extracted from meeting data.

Examples:

- speaking share
- response latency
- interruption
- question detection
- recap behavior
- gain/loss framing
- certainty language

### 5.5 Law Trigger

A system event indicating that the conditions for a behavioral law have been met strongly enough to justify a coaching opportunity.

### 5.6 Intervention

A real-time prompt shown privately to the user.

### 5.7 Prompt Fatigue

The condition where too many prompts reduce usefulness or distract the user.

### 5.8 Explainability Record

A stored trace linking a prompt or post-meeting insight to:

- source events
- extracted features
- triggered laws
- confidence scores

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## 6. Product Vision and Success Criteria

### 6.1 Vision

During a live Google Meet call, the user receives subtle, private prompts that improve listening, clarity, persuasion, decision hygiene, and execution.

### 6.2 Success Metrics

For v1, engineering should instrument:

- prompt latency
- prompt display rate
- prompt suppression rate
- prompt follow-through proxy rate
- post-meeting user satisfaction
- percentage of prompts linked to explainability records
- feature extraction confidence distribution
- meeting completion without technical failure

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# 7. System Context

## 7.1 External Systems

- Google authentication provider
- Google Meet in-browser DOM/runtime environment
- backend API
- storage/database
- model inference services
- transcription or streaming text services where applicable

## 7.2 Internal Logical Components

- browser extension UI
- Google Meet adapter
- consent manager
- event capture layer
- streaming feature extractor
- law engine
- intervention engine
- prompt renderer
- post-meeting summary generator
- user/account service
- privacy and retention service
- audit/explainability service

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# 8. High-Level Architecture

## 8.1 Architecture Overview

The system shall be split into the following logical services/modules:

1. **Extension Client**
    - detects live Google Meet sessions
    - obtains consent
    - renders prompt UI
    - captures supported client-side signals
    - sends normalized events to backend
    - stores minimal ephemeral local state
2. **Meeting Adapter Layer**
    - Google Meet adapter in v1
    - converts platform-specific signals to normalized internal event schema
3. **Streaming Event Pipeline**
    - receives real-time events
    - validates and timestamps them
    - maintains meeting session state
    - forwards to feature extraction
4. **Feature Extraction Engine**
    - derives live rolling features
    - computes confidence scores
    - updates meeting state
5. **Behavioral Law Engine**
    - loads active registry entries
    - evaluates event and rolling-window conditions
    - creates law-trigger candidates
6. **Intervention Engine**
    - ranks candidate prompts
    - applies throttling
    - enforces safety rules
    - emits final prompt decisions
7. **Prompt Delivery Channel**
    - returns prompt payloads to extension
    - renders silent, private UI
8. **Post-Meeting Analysis Service**
    - aggregates meeting evidence
    - generates summary
    - links to live interventions
    - stores report
9. **Identity, Privacy, and Retention Service**
    - authentication
    - user preferences
    - deletion
    - retention management
10. **Registry Service**

- stores law definitions
- versions entries
- activates/deactivates laws

11. **Audit and Explainability Service**

- stores evidence chains
- supports debugging and trust review

### 8.2 Architectural Principle

Core coaching logic must be **platform-agnostic**. Platform-specific capture must be isolated in adapters.

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# 9. Operating Assumptions

- The user explicitly enables coaching for a meeting.
- Google Meet support is implemented via browser extension.
- Not all signals will always be available.
- The system must degrade gracefully under partial observability.
- All coaching must be tied to observable evidence, not broad speculation.

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# 10. Functional Requirements

# 10.1 Authentication and Account

### FR-001

The system shall allow a user to sign up and sign in using Google authentication.

### FR-002

The system shall create a unique user account on first successful authentication.

### FR-003

The system shall associate meetings, preferences, prompts, and reports only to the authenticated user.

### FR-004

The system shall support secure session management for extension and backend API calls.

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## 10.2 Meeting Detection and Session Lifecycle

### FR-005

The extension shall detect when the user enters a Google Meet session.

### FR-006

The extension shall create a local pending meeting context when a Google Meet session is detected.

### FR-007

The system shall not begin data capture until the user explicitly enables coaching for that meeting.

### FR-008

The extension shall show a visible session status indicator with at least these states:

- off
- ready
- active
- muted
- error

### FR-009

The extension shall create a unique `meeting_session_id` when coaching is enabled.

### FR-010

The system shall mark session start time, end time, and duration.

### FR-011

The system shall close the session when the user leaves the meeting, disables coaching, or a session timeout condition is met.

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## 10.3 Consent and Disclosure

### FR-012

The system shall obtain explicit per-meeting user consent before capture begins.

### FR-013

The consent surface shall describe:

- that coaching is private to the user
- what categories of data are captured
- that live prompts may appear during the meeting
- how the user can mute or stop coaching
- how the user can delete meeting data

### FR-014

The system shall persist a `consent_record` for each coached meeting.

### FR-015

The user shall be able to revoke coaching for the current meeting at any time.

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## 10.4 Signal Capture

### FR-016

The Google Meet adapter shall capture all platform-permitted meeting metadata available in-browser.

### FR-017

The system shall support collection of these meeting metadata fields where available:

- meeting_session_id
- platform
- meeting title or label
- start time
- participant count estimate
- duration

- user display identity
- browser version
- extension version

### FR-018

The system shall ingest or derive real-time conversational signals where technically feasible.

### FR-019

The capture layer shall support at minimum these event categories:

- speech started
- speech ended
- turn change
- interruption candidate
- message/transcript segment
- prompt shown
- prompt dismissed
- session state changed

### FR-020

The system shall annotate every event with:

- event_id
- meeting_session_id
- user_id
- event_type
- event_time_utc
- source
- platform
- payload
- capture_confidence if applicable

### FR-021

The system shall support partial operation if transcript text is unavailable.

### FR-022

The system shall support partial operation if only timing and turn-taking signals are available.

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## 10.5 Feature Extraction

### FR-023

The system shall compute streaming features during the meeting.

### FR-024

The system shall support both event-based features and rolling-window features.

### FR-025

The following features shall be implemented in v1 where source data permits:

- `speaking_time_total_seconds`
- `speaking_share_percent`
- `current_continuous_speaking_seconds`
- `turn_count`
- `interruption_count`
- `response_latency_seconds`
- `question_count`
- `clarifying_question_count`
- `summary_or_recap_count`
- `acknowledgment_count`
- `certainty_language_score`
- `hedging_language_score`
- `loss_frame_score`
- `gain_frame_score`
- `action_specificity_score`
- `option_count_presented`
- `default_recommendation_present`
- `owner_assignment_present`
- `deadline_present`
- `evidence_reference_present`
- `peer_example_present`
- `shared_goal_language_present`

### FR-026

The feature extraction engine shall assign a confidence score to each feature update.

### FR-027

The system shall store evidence references sufficient to reconstruct why a feature value was computed.

### FR-028

The system shall maintain rolling windows of at least:

- 30 seconds
- 90 seconds
- full meeting to date

### FR-029

The system shall distinguish direct observations from inferred linguistic classifications.

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## 10.6 Law Registry

### FR-030

The system shall maintain a versioned Behavioral Law Registry.

### FR-031

Each registry entry shall include:

- `law_id`
- `source_family`
- `law_name`
- `status`
- `description`
- `meeting_relevance`
- `observable_inputs`
- `trigger_type`
- `trigger_logic`
- `disconfirming_logic`
- `prompt_templates_live`
- `prompt_templates_post`
- `confidence_threshold`

- `cooldown_seconds`
- `risk_notes`
- `allowed_inferences`
- `version`

### FR-032

The system shall support `status` values:

- draft
- active
- deprecated
- disabled

### FR-033

Only active laws may generate prompts.

### FR-034

The system shall support many-to-many mappings between features and laws.

### FR-035

The system shall support multiple simultaneous trigger candidates.

### FR-036

The system shall log the exact registry version used for each prompt decision.

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## 10.7 Real-Time Law Evaluation

### FR-037

The law engine shall evaluate active laws continuously during a meeting.

### FR-038

The engine shall support two trigger types:

- event

- rolling_window

### FR-039

The engine shall produce a `law_trigger_candidate` when trigger logic is satisfied above threshold.

### FR-040

Each `law_trigger_candidate` shall include:

- trigger_id
- law_id
- meeting_session_id
- event_time_utc
- evidence_refs
- feature_snapshot
- trigger_confidence
- recommended_prompt_type
- expiry_time

### FR-041

The law engine shall suppress a trigger if disconfirming logic is satisfied.

### FR-042

The law engine shall support cooldown at the law level.

### FR-043

The law engine shall not trigger a law if required inputs are missing below a minimum evidence threshold.

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## 10.8 Intervention Engine

### FR-044

The intervention engine shall rank candidate prompts based on:

- trigger confidence
- urgency

- expected usefulness
- novelty
- current prompt fatigue
- timing suitability

### FR-045

The intervention engine shall emit at most one prompt at a time.

### FR-046

The intervention engine shall enforce a global prompt cooldown configurable by user setting.

### FR-047

The intervention engine shall support user-selected coaching intensity levels:

- minimal
- standard
- high-support

### FR-048

The engine shall enforce per-intensity default maximum prompt rates.

Recommended defaults:

- minimal: max 3 prompts per 30 minutes
- standard: max 6 prompts per 30 minutes
- high-support: max 10 prompts per 30 minutes

### FR-049

The intervention engine shall suppress repeated prompts with low novelty.

### FR-050

The intervention engine shall avoid showing prompts while the user is actively speaking unless the prompt is classified as urgent.

### FR-051

The engine shall prefer natural conversational gaps where detectable.

### FR-052

The intervention engine shall classify prompts into:

- pause
- acknowledge
- ask
- frame
- close

### FR-053

Each prompt payload shall include:

- prompt_id
- prompt_type
- short_text
- optional_rationale
- optional_example_phrase
- law_id
- confidence
- expires_at

### FR-054

The system shall log whether a prompt was shown, ignored, muted, or dismissed.

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## 10.9 Prompt UI

### FR-055

The extension shall render prompts privately within the user's browser UI.

### FR-056

Prompt UI shall be silent and visual in v1.

### FR-057

A prompt shall be readable in 1 to 3 seconds.

### FR-058

The default prompt body shall not exceed 12 words.

### FR-059

The optional rationale shall not exceed 10 words.

### FR-060

The optional example phrase shall not exceed 20 words.

### FR-061

The user shall be able to mute prompts for the current meeting.

### FR-062

The user shall be able to disable one or more prompt categories globally.

### FR-063

The UI shall show that coaching is visible only to the user.

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## 10.10 Post-Meeting Summary

### FR-064

The system shall generate a post-meeting report for each coached meeting.

### FR-065

The report shall include:

- meeting metadata
- top strengths
- top growth opportunities
- prompts shown during the meeting
- laws triggered
- key evidence-backed observations
- one to three recommended next actions

### FR-066

The post-meeting report shall distinguish:

- observed facts
- model interpretations
- recommendations

### FR-067

The post-meeting report shall link each major insight to one or more evidence records.

### FR-068

The report shall include a timeline of prompts shown.

### FR-069

The report shall be available from the extension and backend account dashboard.

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## 10.11 History and Trends

### FR-070

The system shall store meeting summaries for historical review unless deleted by the user.

### FR-071

The system shall provide trend views over time for at least:

- listening
- clarity
- concision
- influence
- decision hygiene

### FR-072

The trend view shall show changes over time at a per-user level only.

### FR-073

The system shall not expose any peer comparison in v1.

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## 10.12 User Controls

### FR-074

The user shall be able to configure coaching intensity.

### FR-075

The user shall be able to enable or disable prompt categories.

### FR-076

The user shall be able to choose retention preferences for:

- raw transcript
- derived features
- prompts
- reports

### FR-077

The user shall be able to delete a single meeting and all associated data.

### FR-078

The user shall be able to delete all stored data.

### FR-079

The user shall be able to export their post-meeting reports in machine-readable format.

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# 11. Initial Behavioral Law Registry

## 11.1 Registry Design Constraints

Every law in v1 must satisfy all of the following:

- operationalizable from observable meeting evidence
- safe for real-time prompting
- non-diagnostic
- non-manipulative

- useful in the meeting context
- expressible in short prompt form

# 11.2 Active v1 Laws

## K-01 System 1 Override Risk

**Source Family:** Kahneman
**Law Name:** System 1 vs System 2
**Trigger Type:** event

**Meeting Relevance:** User may react too quickly under pressure.

**Observable Inputs:**

- objection/disagreement detected
- short response latency
- no acknowledgment
- no clarifying question

**Trigger Logic:**
Trigger if:

- disagreement signal present
- response latency < configured threshold
- acknowledgment absent
- clarifying question absent

**Disconfirming Logic:**

- user asks a clarifying question
- user summarizes the other viewpoint
- user acknowledges concern before rebutting

**Allowed Inference:**
User may be responding automatically rather than reflectively.

**Live Prompts:**

- “Pause. Ask one question first.”
- “Acknowledge before rebutting.”
- “Slow down. Clarify first.”

**Cooldown:** 180 seconds
**Confidence Threshold:** 0.72

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## K-02 Loss Aversion Salience

**Source Family:** Kahneman
**Law Name:** Loss Aversion
**Trigger Type:** rolling_window

**Meeting Relevance:** User may be overweighting downside and narrowing the discussion.

**Observable Inputs:**

- repeated loss-framed language
- downside emphasis
- rejection without upside discussion

**Trigger Logic:**
Trigger if loss-framed language materially exceeds balanced tradeoff framing within a rolling window.

**Disconfirming Logic:**

- explicit tradeoff statement
- explicit upside discussion
- balanced risk/opportunity framing

**Allowed Inference:**
User may be over-emphasizing potential loss.

**Live Prompts:**

- “State one upside too.”
- “Frame tradeoffs, not just risks.”
- “Balance downside with opportunity.”

**Cooldown:** 240 seconds
**Confidence Threshold:** 0.70

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## K-03 Early Anchor Introduction

**Source Family:** Kahneman
**Law Name:** Anchoring
**Trigger Type:** event

**Meeting Relevance:** User may be constraining discussion by stating a concrete anchor too early.

**Observable Inputs:**

- early numeric estimate or firm proposal
- low uncertainty language
- no invitation for alternatives

**Trigger Logic:**
Trigger if user introduces an early anchor before exploration and does not mark it as provisional.

**Disconfirming Logic:**

- user labels estimate provisional
- user invites alternatives
- user updates estimate after new evidence

**Allowed Inference:**
User may be unintentionally locking the frame too early.

**Live Prompts:**

- "Mark this as provisional."
- "Invite a second frame."
- "Don't lock the anchor yet."

**Cooldown:** 300 seconds
**Confidence Threshold:** 0.74

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## K-04 Overconfidence Signal

**Source Family:** Kahneman
**Law Name:** Overconfidence
**Trigger Type:** rolling_window

**Meeting Relevance:** User may sound more certain than evidence supports.

**Observable Inputs:**

- repeated certainty markers
- no assumption labeling
- low uncertainty language
- weak evidence citation

**Trigger Logic:**
Trigger if certainty language remains high while evidence references remain low in ambiguous discussion.

**Disconfirming Logic:**

- user names assumptions
- user expresses uncertainty
- user separates fact from estimate

**Allowed Inference:**
User may be signaling excessive certainty.

**Live Prompts:**

- “Name the assumption.”
- “Separate fact from estimate.”
- “Add uncertainty explicitly.”

**Cooldown:** 240 seconds
**Confidence Threshold:** 0.71

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## C-01 Reciprocity Activation

**Source Family:** Cialdini
**Law Name:** Reciprocity
**Trigger Type:** event

**Meeting Relevance:** Acknowledgment first may increase receptivity.

**Observable Inputs:**

- user seeks agreement after pushback
- no acknowledgment
- no concession or recognition
- direct persuasion attempt

**Trigger Logic:**
Trigger when the user tries to persuade after resistance without first recognizing the other side.

**Disconfirming Logic:**

- acknowledgment present
- concession present

- appreciation or recognition present

**Allowed Inference:**
A recognition-first move may improve receptivity.

**Live Prompts:**

- “Acknowledge their point first.”
- “Recognize, then request.”
- “Offer value before asking.”

**Cooldown:** 180 seconds
**Confidence Threshold:** 0.73

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## C-02 Commitment and Consistency

**Source Family:** Cialdini
**Law Name:** Commitment/Consistency
**Trigger Type:** rolling_window

**Meeting Relevance:** Small commitments may move stalled conversations forward.

**Observable Inputs:**

- repeated push for full agreement
- stalled persuasion
- no partial-commitment ask

**Trigger Logic:**
Trigger when conversation is stalled and user continues seeking full buy-in instead of a smaller step.

**Disconfirming Logic:**

- partial agreement ask present
- next-step ask present
- user confirms one agreed point

**Allowed Inference:**
A smaller ask may be more effective.

**Live Prompts:**

- “Ask for a smaller commitment.”
- “Confirm one point of agreement.”

- “Get the next step, not full buy-in.”

**Cooldown:** 240 seconds
**Confidence Threshold:** 0.72

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## C-03 Social Proof Opportunity

**Source Family:** Cialdini
**Law Name:** Social Proof
**Trigger Type:** rolling_window

**Meeting Relevance:** Peer examples may reduce uncertainty about a change.

**Observable Inputs:**

- user is advocating change
- resistance due to unfamiliarity
- no peer or precedent example

**Trigger Logic:**
Trigger when uncertainty-based resistance appears and user has not referenced comparable examples.

**Disconfirming Logic:**

- peer example present
- precedent cited
- normative reference present

**Allowed Inference:**
A comparable example may help.

**Live Prompts:**

- “Use a peer example.”
- “Reference a similar case.”
- “Show this isn’t unusual.”

**Cooldown:** 300 seconds
**Confidence Threshold:** 0.69

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## C-04 Authority Signal

**Source Family:** Cialdini
**Law Name:** Authority
**Trigger Type:** rolling_window

**Meeting Relevance:** Credible sources or data may strengthen recommendations.

**Observable Inputs:**

- user recommendation present
- evidence reference absent
- credibility challenge or hesitation present

**Trigger Logic:**
Trigger when recommendation requires support but authoritative evidence is not cited.

**Disconfirming Logic:**

- evidence cited
- precedent cited
- expert or data reference present

**Allowed Inference:**
Grounding in expertise may strengthen the case.

**Live Prompts:**

- “Cite the evidence.”
- “Name the source.”
- “Ground this in data.”

**Cooldown:** 300 seconds
**Confidence Threshold:** 0.70

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## T-01 Choice Architecture Overload

**Source Family:** Thaler
**Law Name:** Choice Architecture
**Trigger Type:** rolling_window

**Meeting Relevance:** Too many options can increase decision friction.

**Observable Inputs:**

- many options presented
- no recommendation hierarchy

- decision stall

**Trigger Logic:**
Trigger when option count exceeds configured threshold and no narrowing/default exists.

**Disconfirming Logic:**

- options narrowed
- recommendation hierarchy present
- default proposed

**Allowed Inference:**
The choice set may be too complex.

**Live Prompts:**

- “Narrow to two choices.”
- “Structure the options.”
- “Recommend a default.”

**Cooldown:** 300 seconds
**Confidence Threshold:** 0.74

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## T-02 Default Effect Opportunity

**Source Family:** Thaler
**Law Name:** Default Effect
**Trigger Type:** rolling_window

**Meeting Relevance:** A clear default can unblock group decisions.

**Observable Inputs:**

- multiple options remain open
- indecision persists
- no baseline path stated

**Trigger Logic:**
Trigger when the group is stalled and no default path has been proposed.

**Disconfirming Logic:**

- default path named
- recommendation named
- clear baseline already exists

**Allowed Inference:**
A default recommendation may reduce friction.

**Live Prompts:**

- “Propose a default.”
- “Name the baseline option.”
- “Offer the simplest path.”

**Cooldown:** 300 seconds
**Confidence Threshold:** 0.71

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## T-03 Nudge Toward Action

**Source Family:** Thaler
**Law Name:** Nudge
**Trigger Type:** event

**Meeting Relevance:** Small structure cues can convert agreement into action.

**Observable Inputs:**

- agreement language present
- no owner assigned
- no due date
- no explicit next step

**Trigger Logic:**
Trigger near discussion close when agreement exists but action structure is missing.

**Disconfirming Logic:**

- owner present
- next step present
- deadline present

**Allowed Inference:**
A small procedural nudge may improve follow-through.

**Live Prompts:**

- “Assign owner and date.”
- “Convert agreement into next step.”
- “Name the action now.”

**Cooldown:** 180 seconds
**Confidence Threshold:** 0.76

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### T-04 Present Bias Cue

**Source Family:** Thaler
**Law Name:** Present Bias
**Trigger Type:** rolling_window

**Meeting Relevance:** Immediate friction may be getting more attention than future benefit.

**Observable Inputs:**

- immediate effort objections
- no future payoff framing
- short-term cost emphasis

**Trigger Logic:**
Trigger when near-term friction dominates and long-term upside remains unspoken.

**Disconfirming Logic:**

- future payoff stated
- investment framing used
- short-term cost compared to long-term benefit

**Allowed Inference:**
The current framing may overweight immediate inconvenience.

**Live Prompts:**

- “Name the long-term payoff.”
- “Compare cost to future gain.”
- “Reframe this as investment.”

**Cooldown:** 300 seconds
**Confidence Threshold:** 0.68

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# 12. Prompt Taxonomy

## 12.1 Prompt Types

The system shall support exactly these prompt types in v1:

- `pause`
- `acknowledge`
- `ask`
- `frame`
- `close`

### 12.2 Prompt Construction Rules

Each prompt must contain:

- one primary instruction
- optional minimal rationale
- optional example phrase

### 12.3 Prompt Style Rules

Prompts must be:

- brief
- non-judgmental
- behavior-specific
- easy to act on immediately
- free of diagnosis

### 12.4 Forbidden Prompt Styles

The system shall not generate prompts that:

- label the user's personality
- imply pathology
- instruct deception
- instruct coercion
- make unsupported claims about what others think

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## 13. Safety Rules

### SR-001

No prompt may be generated without an active law and supporting evidence.

### SR-002

No prompt may state or imply a stable trait.

### SR-003

No prompt may advise deception, manipulation, or concealment of truth.

### SR-004

No prompt may infer mental health or emotional pathology.

### SR-005

No prompt may be shown if confidence is below the law threshold.

### SR-006

The system shall favor silence over low-confidence coaching.

### SR-007

The system shall clearly separate observed fact from inferred recommendation in post-meeting reports.

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# 14. Non-Functional Requirements

## 14.1 Performance

### NFR-001

The target end-to-end latency from trigger-ready event ingestion to prompt display shall be under 3 seconds for event-based prompts.

### NFR-002

The system should achieve under 2 seconds median latency under normal load.

### NFR-003

Post-meeting reports should be available within 2 minutes after meeting end for meetings under 60 minutes.

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## 14.2 Reliability

### NFR-004

The system shall degrade gracefully when some signals are unavailable.

### NFR-005

The system shall continue providing prompts based on available evidence where safe.

### NFR-006

The system shall not fabricate features when source signals are missing.

### NFR-007

If backend prompting is unavailable, the extension shall continue to show session state and log the failure.

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## 14.3 Privacy

### NFR-008

Privacy shall be a first-class system property.

### NFR-009

The system shall minimize collection of unnecessary third-party data.

### NFR-010

Raw content retention shall default to shorter duration than derived insights.

### NFR-011

The user shall control retention settings.

## 14.4 Security

### NFR-012

All data in transit shall be encrypted.

### NFR-013

Sensitive data at rest shall be encrypted.

### NFR-014

Access to production data shall be least-privilege and auditable.

### NFR-015

Secrets shall be centrally managed and rotated.

---

## 14.5 Explainability

### NFR-016

Every prompt shall be traceable to evidence, features, and laws.

### NFR-017

Every post-meeting insight shall include evidence references.

### NFR-018

Explainability records shall be queryable by engineering/admin tooling with proper authorization.

---

## 14.6 Portability

### NFR-019

Core law evaluation and intervention logic shall be platform-agnostic.

### NFR-020

Adding a new platform shall require only a new adapter plus configuration, not rewriting the law engine.

---

# 15. Data Model

# 15.1 Core Entities

### User

Fields:

- `user_id`
- `google_subject_id`
- `email`
- `display_name`
- `created_at`
- `status`
- `preferences_json`

### MeetingSession

Fields:

- `meeting_session_id`
- `user_id`
- `platform`
- `meeting_label`
- `started_at`
- `ended_at`
- `duration_seconds`
- `extension_version`
- `status`

### ConsentRecord

Fields:

- `consent_record_id`
- `meeting_session_id`
- `user_id`
- `consent_version`
- `granted_at`
- `revoked_at`
- `scope_json`

### RawEvent

Fields:

- `event_id`
- `meeting_session_id`
- `event_type`
- `event_time_utc`
- `source`
- `payload_json`
- `capture_confidence`

### FeatureObservation

Fields:

- `feature_observation_id`
- `meeting_session_id`
- `feature_name`
- `feature_value`
- `window_type`
- `computed_at`
- `confidence`
- `evidence_refs_json`

### LawRegistryEntry

Fields:

- `law_id`
- `version`

- `status`
- `source_family`
- `law_name`
- `definition_json`

### LawTrigger

Fields:

- `trigger_id`
- `meeting_session_id`
- `law_id`
- `law_version`
- `triggered_at`
- `trigger_confidence`
- `evidence_refs_json`
- `feature_snapshot_json`
- `suppressed_reason`

### PromptEvent

Fields:

- `prompt_id`
- `meeting_session_id`
- `law_id`
- `prompt_type`
- `short_text`
- `rationale_text`
- `example_phrase`
- `shown_at`
- `expired_at`
- `display_state`
- `dismissed_at`

### PostMeetingReport

Fields:

- `report_id`
- `meeting_session_id`

- `generated_at`
- `summary_json`
- `insights_json`
- `strengths_json`
- `growth_areas_json`
- `timeline_json`

### DeletionAudit

Fields:

- `deletion_audit_id`
- `user_id`
- `scope`
- `requested_at`
- `completed_at`
- `status`

---

## 15.2 Relationships

- one user has many meeting sessions
- one meeting session has one consent record
- one meeting session has many raw events
- one meeting session has many feature observations
- one meeting session has many law triggers
- one meeting session has many prompts
- one meeting session has one post-meeting report

---

## 16. API Requirements

## 16.1 Client-to-Backend APIs

### POST /auth/session

Establish authenticated session for extension.

### POST /meetings/start

Create meeting session after consent.

Request:

- user token
- platform
- meeting metadata
- extension version

Response:

- meeting_session_id
- session config
- active laws subset
- user preferences

### POST /events/batch

Ingest event batch.

Request:

- meeting_session_id
- array of normalized events

Response:

- accepted count
- errors
- optional prompt payloads

### GET /prompts/poll

Fetch pending prompts for session if polling mode is used.

### POST /prompts/ack

Acknowledge prompt shown/dismissed/muted.

### POST /meetings/end

Close session.

### GET /reports/{meeting_session_id}

Fetch post-meeting report.

### GET /history

Fetch user meeting history.

### DELETE /meetings/{meeting_session_id}

Delete a single meeting.

### DELETE /user/data

Delete all user data.

### GET /registry/active

Fetch active registry entries or hashes for debugging/admin tools.

---

## 16.2 Internal Service APIs

Internal APIs may be event-driven or RPC, but must support:

- feature updates
- law trigger generation
- intervention ranking
- report generation
- retention cleanup

---

# 17. Processing Flows

## 17.1 Live Meeting Flow

1. User enters Google Meet.
2. Extension detects meeting context.
3. User enables coaching.
4. `/meetings/start` is called.
5. Extension begins sending normalized event batches.
6. Feature engine updates meeting state.
7. Law engine evaluates triggers.
8. Intervention engine ranks candidate prompts.
9. Prompt is returned to extension.

10. Extension renders prompt privately.
11. Prompt event is logged.
12. Session continues until meeting ends or coaching is disabled.
13. `/meetings/end` is called.
14. Post-meeting report is generated.

## 17.2 Deletion Flow

1. User requests deletion.
2. System marks relevant records pending deletion.
3. Raw content, feature records, prompts, triggers, and reports are deleted.
4. Deletion audit record is stored.

---

# 18. Privacy and Security Requirements

## 18.1 Privacy Requirements

### PR-001

User coaching outputs shall be private to the user by default.

### PR-002

The system shall not expose user coaching data to employers or other participants.

### PR-003

The system shall provide transparent disclosure of data categories collected.

### PR-004

The system shall support configurable retention periods.

### PR-005

The system shall allow deletion of all user data on demand.

---

## 18.2 Security Requirements

### SEC-001

All API traffic shall use TLS.

### SEC-002

Sensitive stored data shall be encrypted at rest.

### SEC-003

Administrative access to production data shall be logged.

### SEC-004

Production roles shall use least privilege.

### SEC-005

Secrets shall not be embedded in client code.

### SEC-006

The extension shall not store long-lived secrets locally.

---

## 19. Error Handling and Degradation

### ER-001

If the extension cannot detect supported meeting signals, it shall notify the user and avoid false coaching.

### ER-002

If transcript-based features are unavailable, the system shall continue with timing-based features where possible.

### ER-003

If backend is unavailable, the extension shall switch to safe degraded mode and stop prompting.

### ER-004

Prompt failures shall not crash the meeting session.

### ER-005

All critical failures shall be logged with correlation ids.

---

## 20. Observability Requirements

### OBS-001

The system shall log:

- session start/end
- event ingestion counts
- feature update counts
- law trigger counts
- prompt shown counts
- suppression reasons
- report generation status
- API error rates
- prompt latency

### OBS-002

The system shall expose dashboards for:

- prompt latency
- session success rate
- prompt volume per user intensity setting
- trigger-to-prompt conversion rate
- degraded-mode rate

---

## 21. Acceptance Criteria

v1 is acceptable when all of the following are true:

### AC-001

A user can install the extension, sign in, and enable coaching in Google Meet.

### AC-002

The extension visibly shows meeting coach state.

### AC-003

The system captures normalized live events during a meeting after consent.

### AC-004

The feature engine computes streaming features with evidence references.

### AC-005

The law engine evaluates at least the active v1 registry laws.

### AC-006

The intervention engine shows private prompts in real time.

### AC-007

Event-based prompts are displayed within the target latency under normal conditions.

### AC-008

Prompts are throttled and do not spam the user.

### AC-009

Every shown prompt is traceable to a law and evidence chain.

### AC-010

A post-meeting report is generated after the meeting.

### AC-011

The user can mute coaching during a meeting.

### AC-012

The user can delete a meeting and all associated stored data.

---

# 22. Recommended Implementation Plan

# 22.1 Suggested v1 Technical Stack

This is advisory, not mandatory.

### Client

- browser extension: TypeScript
- UI: React
- state: lightweight client store

### Backend

- API: TypeScript/Node or Python/FastAPI
- event processing: async queue or stream
- database: Postgres
- cache/state: Redis
- analytics/logging: standard observability stack

### Registry

- JSON-backed law definitions stored in DB and versioned in source control

### Models

- use deterministic/rule-based trigger logic first
- use ML/LLM only for narrow classification or post-meeting narrative
- never let an LLM be the sole basis for a live prompt decision

# 22.2 Suggested Phased Delivery

### Phase 1

- Google auth
- extension shell
- session detection
- consent
- event transport
- prompt UI

### Phase 2

- timing-based features
- first 4 laws
- prompt throttling
- post-meeting report MVP

### Phase 3

- transcript/language features
- full active registry
- trends
- retention/deletion UI

---

# 23. Developer Notes

## 23.1 Hard Constraints

- Do not overfit to Google Meet DOM internals without abstraction.
- Keep normalized events stable and versioned.
- Keep law definitions declarative where possible.
- Keep evidence storage compact but sufficient for reconstruction.
- Treat silence as preferable to low-confidence prompting.

## 23.2 Coding-Agent Guidance

A coding agent should implement this system in the following order:

1. domain models
2. session lifecycle
3. normalized event schema
4. ingestion API
5. feature extraction
6. law registry loader
7. law evaluator
8. intervention ranking and throttling
9. prompt UI
10. post-meeting reports
11. deletion and retention controls
12. observability

---

## 24. Minimal Event Schema


```JSON
{
  "event_id": "uuid",
  "meeting_session_id": "uuid",
  "user_id": "uuid",
  "platform": "google_meet",
  "event_type": "transcript_segment",
  "event_time_utc": "2026-03-18T15:04:05Z",
  "source": "extension",
  "capture_confidence": 0.88,
  "payload": {
    "text": "I think we should probably go with option A",
    "speaker": "user",
    "start_offset_ms": 151230,
    "end_offset_ms": 153910
  }
}
```


---

## 25. Minimal Law Entry Schema


```JSON
{
  "law_id": "K-01",
  "version": "1.0.0",
  "status": "active",
  "source_family": "Kahneman",
  "law_name": "System 1 vs System 2",
  "description": "Detects rapid rebuttal under disagreement
without acknowledgment or clarification.",
  "meeting_relevance": "Helps the user slow reactive responses.",
  "observable_inputs": [
    "disagreement_detected",
    "response_latency_seconds",
```

```
    "acknowledgment_count",
    "clarifying_question_count"
  ],
  "trigger_type": "event",
  "trigger_logic": {
    "all": [
      {"feature": "disagreement_detected", "op": "eq", "value": true},
      {"feature": "response_latency_seconds", "op": "lt", "value": 1.5},
      {"feature": "acknowledgment_count", "op": "eq", "value": 0},
      {"feature": "clarifying_question_count", "op": "eq", "value": 0}
    ]
  },
  "disconfirming_logic": {
    "any": [
      {"feature": "acknowledgment_count", "op": "gt", "value": 0},
      {"feature": "clarifying_question_count", "op": "gt", "value": 0}
    ]
  },
  "prompt_templates_live": [
    {"type": "ask", "text": "Pause. Ask one question first."},
    {"type": "acknowledge", "text": "Acknowledge before rebutting."}
  ],
  "prompt_templates_post": [
    "You often moved quickly to rebuttal before clarifying the concern."
  ],
  "confidence_threshold": 0.72,
  "cooldown_seconds": 180,
  "risk_notes": [
```

```
    "Do not infer trait impulsivity."
  ],
  "allowed_inferences": [
    "User may be reacting automatically rather than
reflectively."
  ]
}
```

---

## 26. Minimal Prompt Ranking Heuristic

A simple first implementation should score candidate prompts as:

```
score = (trigger_confidence * 0.35) + (urgency * 0.20) + (novelty *
0.15) + (timing_fit * 0.15) + (estimated_usefulness * 0.15) -
fatigue_penalty
```

Use deterministic heuristics first. Refine later.

---

## 27. Open Engineering Decisions

These are expected implementation choices, not blockers:

- exact Google Meet signal-capture method
- polling vs push for prompt delivery
- transcript provider strategy
- where rolling meeting state is stored
- whether linguistic classifiers are rules, models, or hybrid
- UI placement and styling within extension

---

## 28. Final Delivery Definition

A coding agent should treat this SRS as defining a product where:

- the extension runs on Google Meet

- the backend processes real-time events
- features are computed continuously
- a law registry drives prompt eligibility
- prompts are throttled and private
- post-meeting reports summarize what happened
- the user controls consent, retention, and deletion

If you want, I can turn this into a **developer task breakdown with epics, user stories, API contracts, DB schema, and pseudocode for the law engine**.